0 1 2 3 4 5 6 7 8 9

arr → [6, -1, -3, 4, -2, 2, 4, 6, -12, -7]

Sum = 0

Count = 1 2 3 4

$O(n^2)$

```
for (i=0; i<n; i++)
{
    int sum = 0;
    for (j = i; j<n; j++)
    {
        sum += arr[j];
        if (sum == 0)
            count++;
    }
}
```

0 | 1 | 2
0 1 | 1,2 | 2,3
0 1 2 | 1,2,3 | 3,4
0 1 2 3 | |
0, n-1 | 1, n-1 |

pair<int, int> p;

use

$O(n)$

$O(1)$

$O(n)$

0 1 2 3 4 5 6 7 8 9

[6 , -1 , -3 , 4 , -2 , 2 , 4 , 6 , -12 , 7]

Prefix sum ϕ 6 5 2 6 4 6 10 16 4 11

count

(16,1)

(10,1)

(4,2)

(11,1)

result = ~~1~~ ~~3~~ 4

numb — How many time that number came

```
unordered_map<int, int> count;
int result = 0;
int prefixSum = 0;

For (int value : arr)
{
    prefixSum += value;
    result += count[prefixSum];
    Count[prefixSum]++;
}
```

O(N)

(N)

(6,2)

(5,1)

(2,1)

```
if (count.find(prefixSum) != count.end())
{
    result += count[prefixSum];
    count[prefixSum]++;
}
else
{
    count[prefixSum] = 1;
}
```

6

[0, 0, 5, 5, 0, 0]

prefix 0 0

result = 3

initial

{0, 1}

{0, 1}

[1, 2, 3]

prefix 0 1 3 6

result = 1

k = 3

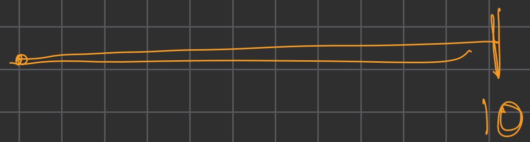
$O(N^2)$

6 - 3

3

3 6

$K=3$



$\boxed{k = 4}$

3	2	2	6	-4	5	-3
3	5	7	13	9	14	11

7

result = ~~1~~

~~2~~

~~3~~

4

3

7

$[4, 5, 0, -2, -3, 1], k = 5$

prefix $\quad 4 \quad 9 \quad 9 \quad 7 \quad 4 \quad 5$

result $= \cancel{1}$

2

result = ~~1~~ ~~2~~ ~~4~~ ~~5~~ 6 ✓

$K = 7$

	0	1	2	3	4	5	6	7	8	9
	2	3	-8	-3	11	4	8	6	9	4
	2	5	-3	-6	5	9	17	23	32	36
reminder	2	5	4	1	5	2	3	2	4	1

$-5 \% 7$

$(-5 + 7) \% 7 \Rightarrow 2$

$-12 \% 7$

$(-5 + 7) \% 7$

$23 - 7 \Rightarrow 16 \% 7 = 2$ — 16

$23 - 14 \Rightarrow 9 \% 7 = 2$ — 9

$23 - 21 + 2 \% 7 = 2$ — 2

$23 - 28 + -5 \% = 2$ — -5

$23 - 35 + -12 = 2$ — -12

-19

-26 $\Rightarrow 6$

$(-1 + 7) \% 7$

$= 6 \% 7$

$-3 \% 7$

$(-3 + 7) \% 7 \Rightarrow 4$

$$\text{reminder} = \left(\left((\text{prefixSum} \% K) + K\right) \% K\right)$$

$9 \% 7 \Rightarrow 2$

$(2 + 2) \% 7$

[10, 5, 2, 6], K = 100

[10] ✓ [10, 5]

[5]

[10, 5]

[10

[5, 2] → [2]

[5, 2]

10

10 60

[5, 2, 6] ✓

[2, 6] ✓

[6] ✓

[5, 2, 6]

result = ~~1~~ ~~2~~ ~~3~~ ~~5~~ 8

K = 10.0

0	1	2	3	4	5
5	4	3	8	5	3

15

result = ~~1~~

~~3~~

~~6~~ ~~9~~ ~~11~~ 13 → 1 prob10